$\rho^\star$ | $N^\star$ | $D^\star$

Reproducing Kaplan et al.

$r = -0.668$ $(-0.69, -0.65)$
$\rho^\star(C_C) = 0.01$ (0.01, 0.01)

$a = 0.835$ $(0.82, 0.85)$
$N^\star(C_C) = 3\text{T}$ (3T, 4T)

$b = 0.163$ $(0.15, 0.18)$
$D^\star(C_C) = 32\text{B}$ (28B, 39B)

Counting last layer FLOPs

$r = -0.402$ $(-0.42, -0.37)$
$\rho^\star(C_C) = 0.18$ (0.13, 0.28)

$a = 0.706$ $(0.69, 0.72)$
$N^\star(C_C) = 787\text{B}$ (630B, 916B)

$b = 0.294$ $(0.28, 0.31)$
$D^\star(C_C) = 124\text{B}$ (109B, 159B)

Correcting warmup

$r = -0.207$ $(-0.24, -0.17)$
$\rho^\star(C_C) = 1.11$ (0.67, 1.85)

$a = 0.602$ $(0.59, 0.62)$
$N^\star(C_C) = 292\text{B}$ (249B, 355B)

$b = 0.398$ $(0.38, 0.41)$
$D^\star(C_C) = 337\text{B}$ (275B, 393B)

Cosine decay (no tuning)

$r = -0.143$ $(-0.18, -0.12)$
$\rho^\star(C_C) = 2.93$ (1.73, 4.24)

$a = 0.571$ $(0.56, 0.59)$
$N^\star(C_C) = 183\text{B}$ (152B, 240B)

$b = 0.428$ $(0.41, 0.44)$
$D^\star(C_C) = 535\text{B}$ (410B, 635B)

Optimizer tuning (no decay)

$r = 0.00594$ $(-0.01, 0.02)$
$\rho^\star(C_C) = 16.51$ (13.03, 19.94)

$a = 0.497$ $(0.49, 0.50)$
$N^\star(C_C) = 77\text{B}$ (70B, 86B)

$b = 0.503$ $(0.50, 0.51)$
$D^\star(C_C) = 1\text{T}$ (1T, 1T)

Compute C [FLOPs]

95\% confidence region
Power law fit
Hoffmann law
Kaplan law
Observations
Warmup period